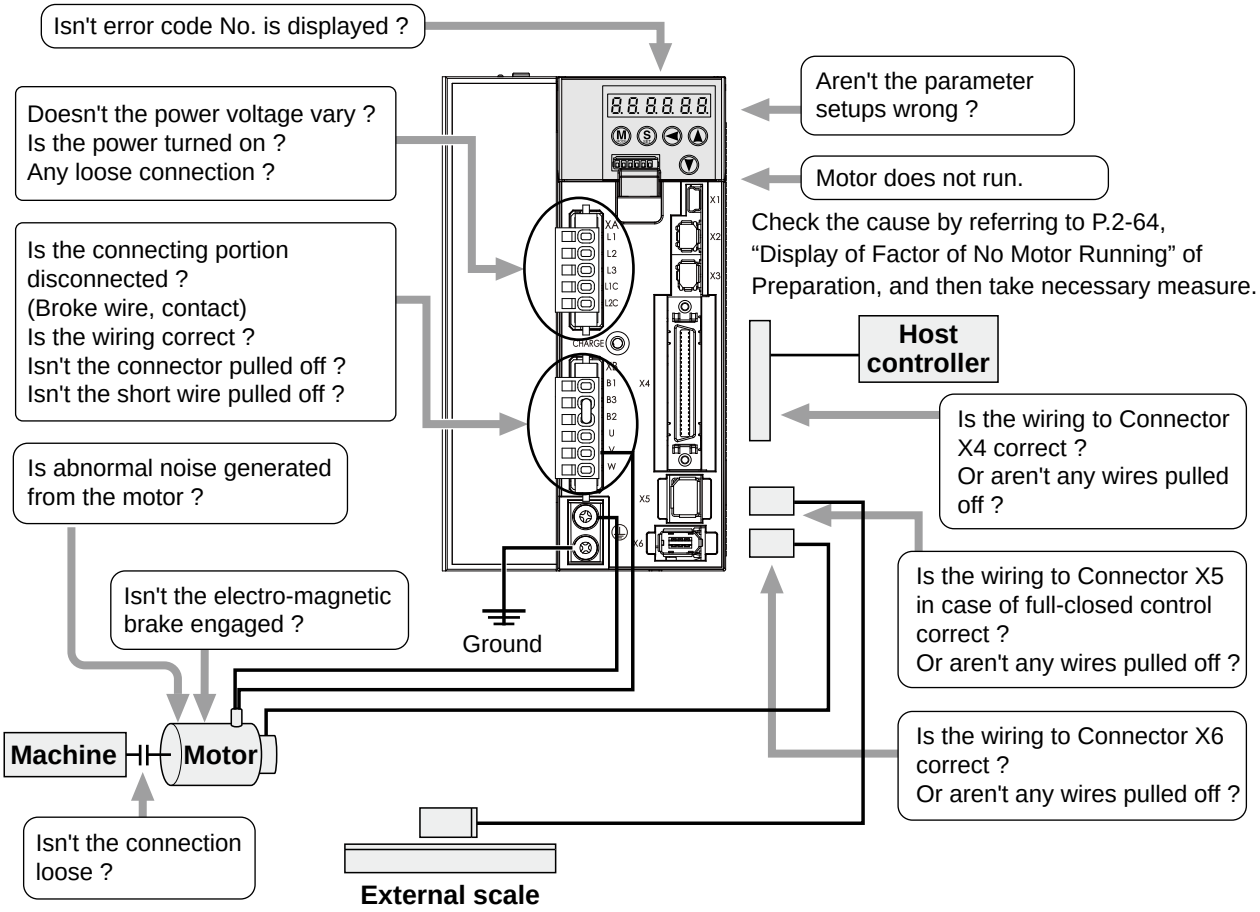
1. When in Trouble

What to Check ?



6 When in Trouble

1. When in Trouble

Protective Function (What is Error Code ?)

- Various protective functions are equipped in the driver. When these are triggered, the motor will stall due to error, the driver will turn the Servo-Alarm output (ALM) to off (open).
- Error status and their measures
  - During the error status, the error code No. will be displayed on the front panel LED, and you cannot turn Servo-ON.
  - You can clear the error status by Alarm clear input(A-CLR) in 120ms or longer.
  - When overload protection is triggered, you can clear it by Alarm clear input(A-CLR) in 10sec or longer after the error occurs. You can clear the Overload protection time characteristics (refer to P.6-13) by turning off the control power supply of the driver.
  - You can clear the above error by operating the front panel keys and setup support softwear "PANATERM".
  - Be sure to clear the alarm during stop after securing safety.

Related page
- P.2-52 ... "How to Use the Front Panel"
- P.3-30 "Inputs and outputs on connector X4"
- P.7-26 "Outline of Setup support software "PANATERM""

<List of error code No.>

Error code		Protective function	Attribute			Detail page
Main	Sub		History	Can be cleared	Immediate stop	
11	0	Control power supply under-voltage protection		○		6-4
12	0	Over-voltage protection	○	○		
13	0	Main power supply under-voltage protection (between P to N)		○		
	1	Main power supply under-voltage protection (AC interception detection)		○		
14	0	Over-current protection	○			6-5
	1	IPM error protection	○			
15	0	Over-heat protection	○		○	
16	0	Over-load protection	○	○		
18	0	Over-regeneration load protection	○		○	6-6
	1	Over-regeneration Tr error protection	○			
21	0	Encoder communication disconnect error protection	○			
	1	Encoder communication error protection	○			
23	0	Encoder communication data error protection	○			
24	0	Position deviation excess protection	○	○	○	6-7
	1	Velocity deviation excess protection	○	○	○	
25	0	Hybrid deviation excess error protection	○		○	
26	0	Over-speed protection	○	○	○	
	1	2nd over-speed protection	○	○		
27	0	Command pulse input frequency error protection	○	○	○	6-8
	2	Command pulse multiplier error protection	○	○	○	
28	0	Limit of pulse replay error protection	○	○	○	
29	0	Deviation counter overflow protection	○	○		
30	0	Safety detection		○		
33	0	IF overlaps allocation error 1 protection	○			6-9
	1	IF overlaps allocation error 2 protection	○			
	2	IF input function number error 1 protection	○			
	3	IF input function number error 2 protection	○			
	4	IF output function number error 1 protection	○			
	5	IF output function number error 2 protection	○			
	6	CL fitting error protection	○			
	7	INH fitting error protection	○			
34	0	Software limit protection	○	○		
36	0 to 2	EEPROM parameter error protection				6-9
37	0 to 2	EEPROM check code error protection				
38	0	Over-travel inhibit input protection		○		
39	0	Analog input1 excess protection	○	○	○	6-10
	1	Analog input2 excess protection	○	○	○	
	2	Analog input3 excess protection	○	○	○	
40	0	Absolute system down error protection	○	○		
41	0	Absolute counter over error protection	○			
42	0	Absolute over-speed error protection	○	○		
43	0	Initialization failure	○			
44	0	Absolute single turn counter error protection	○			
45	0	Absolute multi-turn counter error protection	○			
47	0	Absolute status error protection	○			6-11
48	0	Encoder Z-phase error protection	○			
49	0	Encoder CS signal error protection	○			
50	0	Feedback scale connection error protection	○			
	1	Feedback scale communication error protection	○			
51	0	Feedback scale status 0 error protection	○			
	1	Feedback scale status 1 error protection	○			
	2	Feedback scale status 2 error protection	○			
	3	Feedback scale status 3 error protection	○			
	4	Feedback scale status 4 error protection	○			6-12
	5	Feedback scale status 5 error protection	○			
55	0	A-phase connection error protection	○			
	1	B-phase connection error protection	○			
	2	Z-phase connection error protection	○			
87	0	Compulsory alarm input protection		○		
95	0 to 4	Motor automatic recognition error protection				
Other number		Other error	○			

Note

History...The error will be stored in the error history.
Can be cleared...To cancel the error, use the alarm clear input (A-CLR).
If the alarm clear input is not effective, turn off power, remove the cause of the error and then turn on power again.
Immediate stop...Instantaneous controlled stop upon occurrence of an error.
(Setting of “Pr.5.10 Sequence at alarm” is also required.)

1. When in Trouble

Protective function (Detail of error code)

Protective function	Error code No. Main	Error code No. Sub	Causes	Measures
Control power supply under-voltage protection	**11**	**0**	Voltage between P and N of the converter portion of the control power supply has fallen below the specified value. 1) Power supply voltage is low. Instantaneous power failure has occurred 2) Lack of power capacity...Power supply voltage has fallen down due to inrush current at the main power-on. 3) Failure of servo driver (failure of the circuit)	Measure the voltage between lines of connector and terminal block (L1C - L2C) . 1) Increase the power capacity. Change the power supply. 2) Increase the power capacity. 3) Replace the driver with a new one.
Over-voltage protection	**12**	**0**	Voltage between P and N of the converter portion of the control power supply has exceeded the specified value 1) Power supply voltage has exceeded the permissible input voltage. Voltage surge due to the phase-advancing capacitor or UPS (Uninterruptible Power Supply) have occurred. 2) Disconnection of the regeneration discharge resistor 3) External regeneration discharge resistor is not appropriate and could not absorb the regeneration energy. 4) Failure of servo driver (failure of the circuit)	Measure the voltage between lines of connector (L1, L2 and L3). 1) Enter correct voltage. Remove a phase-advancing capacitor. 2) Measure the resistance of the external resistor connected between terminal P and B of the driver. Replace the external resistor if the value is ∞. 3) Change to the one with specified resistance and wattage. 4) Replace the driver with a new one.
Main power supply under-voltage protection (PN)	**13**	**0**	Instantaneous power failure has occurred between L1 and L3 for longer period than the preset time with Pr6D (Main power off detecting time) while Pr65 (LV trip selection at the main power-off) is set to 1. Or the voltage between P and N of the converter portion of the main power supply has fallen below the specified value during Servo-ON. 1) Power supply voltage is low. Instantaneous power failure has occurred 2) Instantaneous power failure has occurred. 3) Lack of power capacity...Power supply voltage has fallen down due to inrush current at the main power-on. 4) Phase lack...3-phase input driver has been operated with single phase input. 5) Failure of servo driver (failure of the circuit)	Measure the voltage between lines of connector (L1, L2 and L3). 1) Increase the power capacity. Change the power supply. Remove the causes of the shutdown of the magnetic contactor or the main power supply, then re-enter the power. 2) Set up the longer time to Pr5.09 (Main power off detecting time). Set up each phase of the power correctly. 3) Increase the power capacity. For the capacity, refer to P.2-6, "Driver and List of Applicable Peripheral Equipments" of Preparation. 4) Connect each phase of the power supply (L1, L2 and L3) correctly. For single phase, 100V and 200V driver, use L1 and L3. 5) Replace the driver with a new one.
Main power supply under-voltage protection (AC)		**1**		

Protective function	Error code No. Main	Error code No. Sub	Causes	Measures
* **Over-current protection**	**14**	**0**	Current through the converter portion has exceeded the specified value. 1) Failure of servo driver (failure of the circuit, IGBT or other components)	1) Turn to Servo-ON, while disconnecting the motor. If error occurs immediately, replace with a new driver.
* **IPM error protection**		**1**	2) Short of the motor wire (U, V and W)	2) Check that the motor wire (U, V and W) is not shorted, and check the branched out wire out of the connector. Make a correct wiring connection.
			3) Earth fault of the motor wire	3) Measure the insulation resistance between motor wires, U, V and W and earth wire. In case of poor insulation, replace the motor.
			4) Burnout of the motor	4) Check the balance of resister between each motor line, and if unbalance is found, replace the motor.
			5) Poor contact of the motor wire.	5) Check the loose connectors. If they are, or pulled out, fix them securely.
			6) Timing of pulse input is same as or earlier than Servo-ON.	6) Enter the pulses 100ms or longer after Servo-ON.
* **Over-heat protection**	**15**	**0**	Temperature of the heat sink or power device has been risen over the specified temperature. 1) Ambient temperature has risen over the specified temperature. 2) Over-load	1) Improve the ambient temperature and cooling condition. 2) Increase the capacity of the driver and motor. Set up longer acceleration/ deceleration time. Lower the load.
Over-load protection	**16**	**0**	Torque command value has exceeded the over-load level set with Pr5.12 (Setup of over-load level) and resulted in overload protection according to the time characteristics (described later) 1) Load was heavy and actual torque has exceeded the rated torque and kept running for a long time. 2) Oscillation and hunching action due to poor adjustment. Motor vibration, abnormal noise. Inertia ratio (Pr0.04) setup error. 3) Miswiring, disconnection of the motor. 4) Machine has collided or the load has gotten heavy. Machine has been distorted. 5) Electromagnetic brake has been kept engaged. 6) While wiring multiple axes, miswiring has occurred by connecting the motor cable to other axis.	Check that the torque (current) does not oscillates nor fluctuate up an down very much on the graphic screen of the PANATERM. Check the over-load alarm display and load factor with the PANATERM. 1) Increase the capacity of the driver and motor. Set up longer acceleration/ deceleration time. Lower the load. 2) Make a re-adjustment. 3) Make a wiring as per the wiring diagram. Replace the cables. 4) Remove the cause of distortion. Lower the load. 5) Measure the voltage between brake terminals. Release the brake 6) Make a correct wiring by matching the correct motor and encoder wires.

• The over-load protection time characteristics are described on P.6-13.

Caution Once this error occurs, it cannot be cleared at least for 10 sec.

Note

- When protective function marked with * in the protective function table is activated, it cannot be disabled by the alarm clear input (A-CLR). To return to the normal operation, turn off power, remove the cause, and then turn on power again.

Related page • P.2-8 “System Configuration and Wiring” • P.4-44 “Details of parameter”

Protective function	Error code No. Main	Error code No. Sub	Causes	Measures
* **Over-regeneration load protection**	**18**	**0**	Regenerative energy has exceeded the capacity of regenerative resistor. 1) Due to the regenerative energy during deceleration caused by a large load inertia, converter voltage has risen, and the voltage is risen further due to the lack of capacity of absorbing this energy of the regeneration discharge resistor. 2) Regenerative energy has not been absorbed in the specified time due to a high motor rotational speed. 3) Active limit of the external regenerative resistor has been limited to 10% duty.	Check the load factor of the regenerative resistor from the front panel or via communication. Do not use in the continuous regenerative brake application. 1) Check the running pattern (velocity monitor). Check the load factor of the regenerative resistor and over-regeneration warning display. Increase the capacity of the driver and the motor, and loosen the deceleration time. Use the external regenerative resistor. 2) Check the running pattern (speed monitor). Check the load factor of the regenerative resistor. Increase the capacity of the driver and the motor, and loosen the deceleration time. Lower the motor rotational speed. Use an external regenerative resistor. 3) Set up Pr0.16 to 2. **Caution** Install an external protection such as thermal fuse without fail when you set up Pr0.16 to 2. Otherwise, regenerative resistor loses the protection and it may be heated up extremely and may burn out.
* **Regenerative transistor error protection**	**18**	**1**	Regenerative driver transistor on the servo driver is defective.	Replace the driver.
* **Encoder communication disconnection error protection**	**21**	**0**	Communication between the encoder and the driver has been interrupted in certain times, and disconnection detecting function has been triggered.	Make a wiring connection of the encoder as per the wiring diagram. Correct the miswiring of the connector pins.
* **Encoder communication error protection**		**1**	Communication error has occurred in data from the encoder. Mainly data error due to noise. Encoder cables are connected, but communication data has some errors.	• Secure the power supply for the encoder of DC5V±5% (4.75 to 5.25V)...pay an attention especially when the encoder cables are long. • Separate the encoder cable and the motor cable if they are bound together. • Connect the shield to FG.
* **Position deviation excess protection**	**23**	**0**	Data communication between the encoder is normal, but contents of data are not correct. Mainly data error due to noise. Encoder cables are connected, but communication data has some errors.	

Note
- When protective function marked with * in the protective function table is activated, it cannot be disabled by the alarm clear input (A-CLR). To return to the normal operation, turn off power, remove the cause, and then turn on power again.

Related page • P.4-6... “Details of parameter”

Protective function	Error code No. Main	Sub	Causes	Measures
Position deviation excess protection	**24**	**0**	Deviation pulses have exceeded the setup of Pr0.14. 1) The motor movement has not followed the command. 2) Setup value of Pr0.14 (Position deviation excess setup) is small.	1) Check that the motor follows to the position command pulses. Check that the output toque has not saturated in torque monitor. Make a gain adjustment. Set up maximum value to Pr0.13 and Pr5.22. Make a encoder wiring as per the wiring diagram. Set up the longer acceleration/deceleration time. Lower the load and speed. 2) Set up a larger value to Pr0.14.
Velocity deviation excess protection		**1**	The difference between the internal positional command speed and actual speed (speed deviation) exceeds the setup vale of Pr6.02. Note) If the internal positional command speed is forcibly set to 0 due to instantaneous stop caused by the command pulse inhibit input (INH) or CW/CCW over-travel inhibit input, the speed deviation rapidly increases at this moment. Pr6.02 setup value should have sufficient margin because the speed deviation also largely increases on the rising edge of the internal positional command speed.	• Increase the setup value of Pr6.02. • Lengthen the acceleration/deceleration time of internal positional command speed, or improve the follow-up characteristic by adjusting the gain. • Disable the excess speed deviation detection (Pr6.02 = 0).
*** Hybrid deviation excess error protection**	**25**	**0**	Position of load by the external scale and position of the motor by the encoder slips larger than the setup pulses with Pr7B (Setup of hybrid deviation excess) at full-closed control.	• Check the connection between the motor and the load. • Check the connection between the external scale and the driver. • Check that the variation of the motor position (encoder feedback value) and the load position (external scale feedback value) is the same sign when you move the load. Check that the numerator and denominator of the external scale division (Pr3.24 and 3.25) and reversal of external scale direction (Pr3.26) are correctly set.
Over-speed protection	**26**	**0**	The motor rotational speed has exceeded the setup value of Pr5.13.	• Do not give an excessive speed command. • Check the command pulse input frequency and division/multiplication ratio. • Make a gain adjustment when an overshoot has occurred due to a poor gain adjustment. • Make a wiring connection of the encoder as per the wiring diagram.
2nd Over-speed protection		**1**	The motor rotational speed has exceeded the setup value of Pr6.15.	

Protective function	Error code No. Main	Error code No. Sub	Causes	Measures
Command pulse input frequency error protection	**27**	**0**	The frequency of command pulse input is more than 1.2 times the setting in Pr5.32.	Check the command pulse input for frequency.
Electronic gear error protection		**2**	Division and multiplication ratio which are set up with the command pulse counts per single turn and the1st and the 4th numerator/denominator of the electronic gear are not appropriate.	Check the setup value of electronic gear.
Pulse regeneration limit protection	**28**	**0**	The output frequency of pulse regeneration has exceeded the limit.	• Check the setup values of Pr0.11 and 5.03. • To disable the detection, set Pr5.33 to 0.
Deviation counter overflow protection	**29**	**0**	Positional deviation of encoder pulse reference has exceeded 2^{29} (536870912).	• Check that the motor runs as per the position command pulses. • Check that the output toque has not saturated in torque monitor. • Make a gain adjustment. • Set up maximum value to Pr0.13 and Pr5.22. • Make a wiring connection of the encoder as per the wiring diagram.
Safety input protection	**30**	**0**	Input photocoupler of both or one of safety input 1 and 2 is OFF.	Check wiring of safety input 1 and 2.
*** I/F input duplicated allocation error 1 protection**	**33**	**0**	Input signals (SI1, SI2, SI3, SI4, SI5) are assigned with two functions.	Allocate correct function to each connector pin.
*** I/F input duplicated allocation error 2 protection**		**1**	Input signals (SI6, SI7, SI8, SI9, SI10) are assigned with two functions.	
*** I/F input function number error 1 protection**		**2**	Input signals (SI1, SI2, SI3, SI4, SI5) are assigned with undefined number.	
*** I/F input function number error 2 protection**		**3**	Input signals (SI6, SI7, SI8, SI9, SI10) are assigned with undefined number.	

Protective function	Error code No. Main	Error code No. Sub	Causes	Measures
* **I/F output function number error 1 protection**	**33**	**4**	Output signals (SI1, SI2, SI3) are assigned with undefined number.	Allocate correct function to each connector pin.
* **I/F output function number error 2 protection**		**5**	Output signals (SI4) are assigned with undefined number.	
* **CL assignment error**		**6**	Counter clear function is assigned to a signal number other than SI7.	
* **INH assignment error**		**7**	Command pulse inhibit input function is assigned to a signal number other than SI10.	
Motor working range setup error protection	**34**	**0**	When a position command within the specified input range is given, the motor operates outside its working range specified in Pr5.14. 1) Gain is not appropriate. 2) Pr5.14 setup value is low.	1) Check the gain (balance between position loop gain and speed loop gain) and inertia ratio. 2) Increase the setup value of Pr5.14. Or, Set Pr5.14 to 0 to disable the protective function.
* **EEPROM parameter error protection**	**36**	**0** **1** **2**	Data in parameter storage area has been damaged when reading the data from EEPROM at power-on.	• Set up all parameters again. • If the error persists, replace the driver (it may be a failure.) Return the product to the dealer or manufacturer.
* **EEPROM check code error protection**	**37**	**0** **1** **2**	Data for writing confirmation to EEPROM has been damaged when reading the data from EEPROM at power-on.	Replace the driver. (it may be a failure). Return the product to a dealer or manufacturer.
* **Over-travel inhibit input protection**	**38**	**0**	With Pr5.04, over-travel inhibit input setup = 0, both positive and negative over-travel inhibit inputs (POT/NOT) have been ON. With Pr5.04 = 0, positive or negative over-travel inhibit input has turned ON.	Check that there are not any errors in switches, wires or power supply which are connected to positive direction/ negative direction over-travel inhibit input. Check that the rising time of the control power supply (DC12 to 24V) is not slow.

Note

- When protective function marked with * in the protective function table is activated, it cannot be disabled by the alarm clear input (A-CLR). To return to the normal operation, turn off power, remove the cause, and then turn on power again.

Protective function	Error code No. Main	Error code No. Sub	Causes	Measures
Analog input 1 (AI1) excess protection	**39**	**0**	Higher voltage has been applied to the analog input 1 than the value that has been set by Pr4.24.	• Set up Pr4.24 correctly. Check the connecting condition of the connector X4. • Set up Pr4.24 to 0 and invalidate the protective function.
Analog input 2 (AI2) excess protection		**1**	Higher voltage has been applied to the analog input 1 than the value that has been set by Pr4.27.	• Set up Pr4.27 correctly. Check the connecting condition of the connector X4. • Set up Pr4.27 to 0 and invalidate the protective function.
Analog input 3 (AI3) excess protection	**39**	**2**	Higher voltage has been applied to the analog input 1 than the value that has been set by Pr4.30.	• Set up Pr4.30 correctly. Check the connecting condition of the connector X4. • Set up Pr4.30 to 0 and invalidate the protective function.
Absolute system down error protection	**40**	**0**	Voltage of the built-in capacitor has fallen below the specified value because the power supply or battery for the 17-bit absolute encoder has been down. **Caution** Once this error occurs, the alarm cannot be cleared until the absolute encoder is reset.	After connecting the power supply for the battery, clear the absolute encoder.
* **Absolute counter over error protection**	**41**	**0**	Multi-turn counter of the 17-bit absolute encoder has exceeded the specified value.	• Set Pr0.15 to 2 to ignore the multi-turn counter over. • Limit the travel from the machine origin within 32767 revolutions.
Absolute over-speed error protection	**42**	**0**	The motor speed has exceeded the specified value when only the supply from the battery has been supplied to 17-bit encoder during the power failure. **Caution** Once this error occurs, the alarm cannot be cleared until the absolute encoder is reset.	• Check the supply voltage at the encoder side (5V±5%) • Check the connecting condition of the connector, CN X2.
* **Encoder initialization error protection *1**	**43**	**0**	Encoder initialization error was detected.	Replace the motor.
* **Absolute single turn counter error protection *1**	**44**	**0**	Single turn counter error of encoder has been detected.	Replace the motor.
* **Absolute multi-turn counter error protection *1**	**45**	**0**	Multi turn counter error of encoder has been detected.	Replace the motor.

Protective function	Error code No. Main	Error code No. Sub	Causes	Measures
* **Absolute status error protection *1**	**47**	**0**	Encoder has been running at faster speed than the specified value at power-on.	Arrange so as the motor does not run at power-on.
* **Encoder Z-phase error protection*1**	**48**	**0**	Missing pulse of Z-phase of serial incremental encoder has been detected. The encoder might be a failure.	Replace the motor.
* **Encoder CS signal error protection*1**	**49**	**0**	CS signal logic error of serial incremental encoder has been detected. The encoder might be a failure.	Replace the motor.
* **Feedback scale wiring error protection**	**50**	**0**	Communication between the external scale and the driver has been interrupted in certain times, and disconnection detecting function has been triggered.	• Make a wiring connection of the external scale as per the wiring diagram. • Correct the miswiring of the connector pins.
* **External communi-cation data error protection**		**1**	Communication error has occurred in data from the external scale. Mainly data error due to noise. External scale cables are connected, but communication date has some error.	• Secure the power supply for the external scale of DC5±5% (4.75 to 5.25V)...pay attention especially when the external scale cables are long. • Separate the external scale cable and the motor cable if they are bound together. • Connect the shield to FG...refer to wiring diagram.
* **External scale status 0 error protection *1**	**51**	**0**	Bit 0 of the external scale error code (ALMC) has been turned to 1. Check the specifications of the external scale.	Remove the causes of the error, then clear the external scale error from the front panel. And then, shut off the power to reset.
* **External scale status 1 error protection *1**		**1**	Bit 1 of the external scale error code (ALMC) has been turned to 1. Check the specifications of the external scale.	
* **External scale status 2 error protection *1**		**2**	Bit 2 of the external scale error code (ALMC) has been turned to 1. Check the specifications of the external scale.	
* **External scale status 3 error protection *1**		**3**	Bit 3 of the external scale error code (ALMC) has been turned to 1. Check the specifications of the external scale.	

Note

- When protective function marked with * in the protective function table is activated, it cannot be disabled by the alarm clear input (A-CLR). To return to the normal operation, turn off power, remove the cause, and then turn on power again.

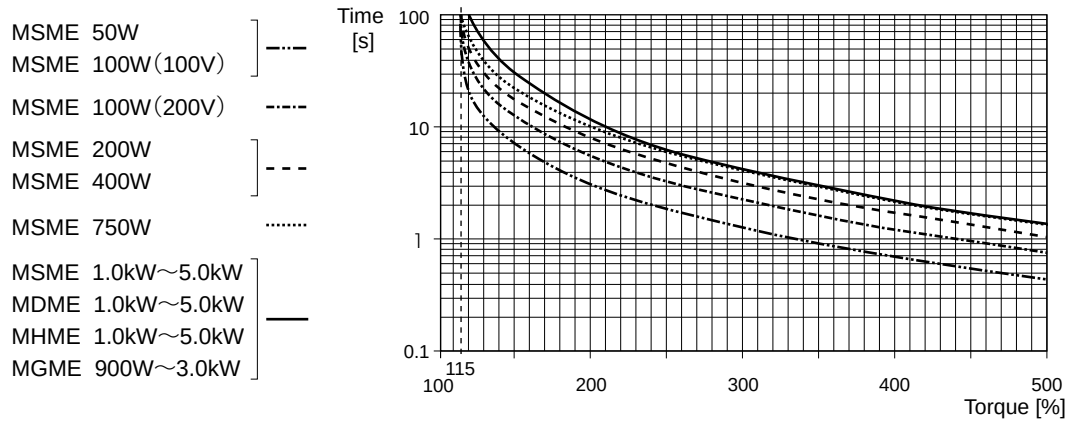
Related page • P.4.6... "Details of parameter"

Protective function	Error code No. Main	Error code No. Sub	Causes	Measures
* **External scale status 4 error protection *1**	**51**	**4**	Bit 4 of the external scale error code (ALMC) has been turned to 1. Check the specifications of the external scale.	Remove the causes of the error, then clear the external scale error from the front panel. And then, shut off the power to reset.
* **External scale status 5 error protection *1**		**5**	Bit 5 of the external scale error code (ALMC) has been turned to 1. Check the specifications of the external scale.	
* **A-phase wiring error protection**	**55**	**0**	A-phase wiring in the external scale is defective, e.g. discontinued.	Check the A-phase wiring connection.
* **B-phase wiring error protection**		**1**	B-phase wiring in the external scale is defective, e.g. discontinued.	Check the B-phase wiring connection.
* **Z-phase wiring error protection**		**2**	Z-phase wiring in the external scale is defective, e.g. discontinued.	Check the Z-phase wiring connection.
Forced alarm input protection	**87**	**0**	Forced alarm input (E-STOP) is applied.	Check the wiring of forced alarm input (E-STOP).
* **Motor automatic recognition error protection**	**95**	**0 to 4**	The motor and the driver has not been matched.	Replace the motor which matches to the driver.
* **Other error**	**Other No.**		Control circuit has malfunctioned due to excess noise or other causes. Some error has occurred inside of the driver while triggering self-diagnosis function of the driver.	• Turn off the power once, then re-enter. • If error repeats, this might be a failure. Stop using the products, and replace the motor and the driver. Return the products to the dealer or manufacturer.

Note

- When protective function marked with * in the protective function table is activated, it cannot be disabled by the alarm clear input (A-CLR). To return to the normal operation, turn off power, remove the cause, and then turn on power again.

Time characteristics of Error code No.16.0 (Overload protection)



Caution Use the motor so that actual torque stays in the continuous running range shown in "S-T characteristic" of the motor. For the S-T characteristics, see P.7-56 Motor characteristics (S-T characteristics).

Software Limit Function (Error code No.34.0)

1) Outline

You can make an alarm stop of the motor with software limit protection (Error code No.34) when the motor travels exceeding the movable range which is set up with Pr5.14 (Set up of software limit) against the position command input range.

You can prevent the work from colliding to the machine end caused by motor oscillation.

2) Applicable range

This function works under the following conditions.

	Conditions under which the software limit works
Control mode	• Position control, Full-closed control
Others	• Should be in servo-on condition • Input signals such as the deviation counter clear and command input inhibit, and parameters except for controls such as torque limit setup, are correctly set, assuring that the motor can run smoothly.

3) Cautions

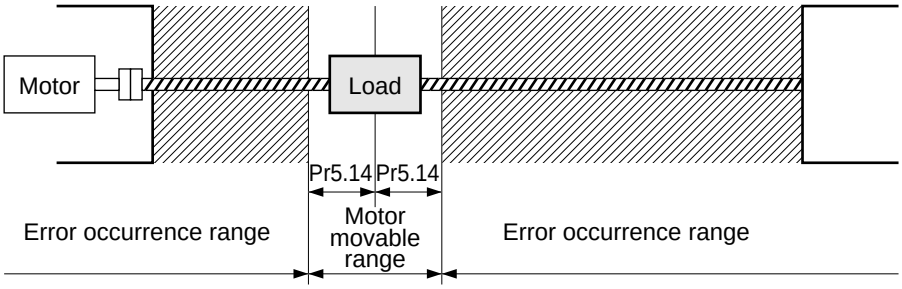
- This function is not a protection against the abnormal position command.
- When this software limit protection is activated, the motor decelerates and stops according to Pr5.14 (Sequence at alarm).
  The work (load) may collide to the machine end and be damaged depending on the load during this deceleration, hence set up the range of Pr5.14 including the deceleration movement.
- This software limit protection will be invalidated during the trial run and frequency characteristics functioning of the PANATERM.

Related page • P.4-44, 4-45 "Details of parameter"

4) Example of movement

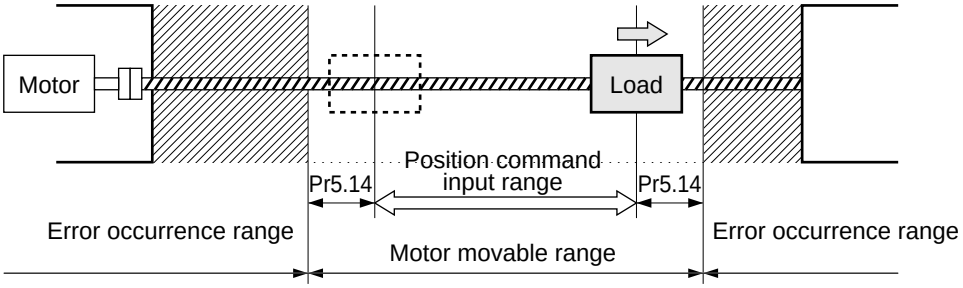
(1) When no position command is entered (Servo-ON status),

The motor movable range will be the travel range which is set at both sides of the motor with Pr26 since no position command is entered. When the load enters to the Err34 occurrence range (oblique line range), software limit protection will be activated.



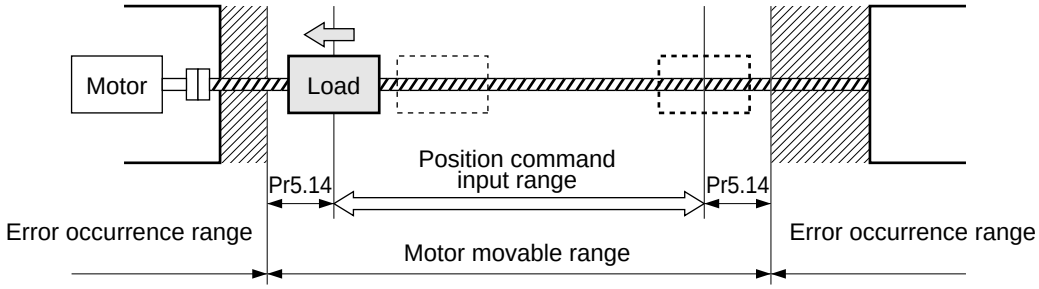
(2) When the load moves to the right (at Servo-ON),

When the position command to the right direction is entered, the motor movable range will be expanded by entered position command, and the movable range will be the position command input range + Pr26 setups in both sides.



(3) When the load moves to the left (at Servo-ON),

When the position command to the left direction, the motor movable range will be expanded further.



5) Condition under which the position command input range is cleared

The position command input range will be 0-cleared under the following conditions.

- when the power is turned on.
- while the position deviation is being cleared (Deviation counter clear is valid, Pr66 (Sequence at over-travel inhibition) is 2 and over-travel inhibition input is valid.)
- At the beginning and ending of trial run via communication.

Related page • P.4-42... “Details of parameter”

2. Setup of gain pre-adjustment protection

Before starting gain adjustment, set the following parameters based on the conditions of use, to assure safe operation.

1) Setup of over-travel inhibit input

By inputting the limit sensor signal to the driver, the bumping against mechanical end can be prevented. Refer to interface specification, positive/negative direction over-travel inhibit input (POT/NOT). Set the following parameters which are related to over-travel inhibit input.

Pr5.04 Setup of over-travel inhibit input

Pr5.05 Sequence at over-travel inhibit

Related page P.3-38 (POT/NOT), P.4-42 (Pr5.04, Pr5.05)

2) Setup of torque limit

By limiting motor maximum torque, damage caused by failure or disturbance such as bite of the machine and collision will be minimized. To apply standardized limit through parameters, set Pr0.13 The 1st torque limit.

If the torque limit setup is lower than the value required during the actual application, the following two protective features will be triggered: over-speed protection when overshoot occurs, and excess positional deviation protection when response to the command delays.

By allocating the torque in-limit output (TLC) of interface specification to the output signal, torque limit condition can be detected externally.

Related page P.3-45 (TLC), P.4-10 (Pr0.13), P.4-47 (Pr5.21)

3) Setup of over-speed protection

Generates Err26.0 Over-speed protection when the motor speed is excessively high.

Default setting is the applicable motor maximum speed [r/min] × 1.2.

If your application operates below the motor maximum speed, set Pr5.13 Setup of over-speed level by using the formula below.

Pr5.13 Setup of over-speed level = Vmax × (1.2 to 1.5)
Vmax: motor maximum speed [r/min] in operating condition
Factor in ( ) is margin to prevent frequent activation of over-speed protection.

When running the motor at a low speed during initial adjustment stage, setup the over-speed protection by multiplying the adjusting speed by a certain margin to protect the motor against possible oscillation.

Related page P.4-45 (Pr5.13)

(Continued ...)

4) Setup of the excess positional deviation protection

During the position control or full-closed control, this function detects potential excessive difference between the positional command and motor position and issues Err24.0 Excess positional deviation protection.

Excess positional deviation level can be set to Pr0.14 Setup of positional deviation excess. The deviation can be detected through command positional deviation [pulse (command unit)] and encoder positional deviation [pulse (encoder unit)], and one of which can be selected by Pr5.20 Position setup unit select. (See the control block diagram.)

Default setting is 100000[pulse (command unit)].

Because the positional deviation during normal operation depends on the operating speed and gain setting, fill the equation below based on your operating condition and input the resulting value to Pr0.14.

• When Pr5.20 = 0 (detection through command positional deviation)

Pr0.14 Setup of positional deviation excess = Vc/kp × (1.2 to 2.0)
Vc: maximum frequency of positional command pulse [pulse (command unit)/s]
Kp: position loop gain [1/s]
Factor in ( ) is margin to prevent frequent activation of excess positional deviation protection.

Note 1) When switching position loop gain Kp, select the smallest value for calculation.
Note 2) When using the positional command filter and damping control, add the following values.
Positional command smoothing filter: Vc × filter time constant [s]
Positional command FIR filter: Vc × filter time constant [s]/2
Damping control: Vc/(π × damping frequency [Hz])

• When Pr5.20 = 1 (detection through encoder positional deviation, full-closed positional deviation)

Pr0.14 Setup of positional deviation excess = Ve/Kp × (1.2 to 2.0)
Ve: maximum operation frequency [pulse/s] in encoder unit or full-closed unit
Kp: position loop gain [1/s]

Note 3) When switching position loop gain Kp, select the smallest value for calculation.
Note 4) When Pr5.20 = 1, setups of positional command filter and damping control have no effect.

Related page P.4-10 (Pr0.14), P4-47 (Pr5.20)

5) Setup of motor working range

During the position control or full-closed control, this function detects the motor position which exceeds the revolutions set to Pr5.14 Motor working range setup, and issues Err34.0 Motor working range protection.

Related page P.4-15 (Pr5.14)

6) Setup of hybrid deviation excess error protection

At the initial operation with full-closed control, operation failure may occur due to reverse connection of external scale or wrong external scale division ratio.

To indicate this type of defect, Err25.0 Hybrid deviation excess error protection is issued when the deviation of motor position (encoder unit) and load position (external scale unit) exceed Pr3.28 Setup of hybrid deviation excess.

Default setting is 16000 pulse (command unit). Because the deviation in normal operation varies with the operation speed and gain setup. Add a margin to this setting according to your operating condition.

Related page P.4-31 (Pr3.28)